Weil Conjectures

Chapter 3: Constructible complexes

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3 Constructible complexes

A lisse E-sheaf is the analogue of a local system: pro-étale locally it is isomorphic to E_X^r for some r (Definition 3.2 below). Inside such a sheaf L, a lattice is a locally free $\mathcal{O}_E$-subsheaf L_0 with $L_0[1/\ell] = L$, and the first point of this chapter is that a lattice need not exist. For an example, glue 0 and ∞ on $\mathbf{P}^1$ to obtain a nodal curve X. Then the infinite chain Y of projective lines, with ∞ on each line glued to 0 on the next, maps to X by an étale covering. Moreover, the group $\mathbf{Z}$ acts on Y by translation, simply transitively on the fibres, so $Y \to X$ is an étale $\mathbf{Z}$-torsor. Using this torsor and letting the generator of $\mathbf{Z}$ act on E by multiplication by ℓ, we obtain a lisse E-sheaf of rank one. It has no lattice, because ℓ is not a unit of $\mathcal{O}_E$, so that multiplication by ℓ does not carry any lattice onto itself. The sheaf comes from the character $n \mapsto \ell^n$ of $\mathbf{Z}$, and since this character does not extend continuously to $\widehat{\mathbf{Z}}$, the sheaf has no classical counterpart as a system (F_n) (see the discussion before Lemma 3.24; the example is in [BS15]).

This matters because every geometric input to the next chapters (proper base change, the cohomology of curves, Deligne's generic base change) is a theorem about torsion sheaves, whereas a rational sheaf has no reduction modulo ℓ: since E_X is ℓ-divisible, we have $E_X \otimes^{\mathbf{L}} \mathcal{O}_E/\pi_E = 0$. To use the torsion theory we must therefore find integral objects inside rational ones, and then pass to their torsion reductions. The example above shows that this cannot be done globally.

Classically each coefficient ring comes with its own theory:

coefficients	classical constructible objects
finite ring A	constructible étale sheaves of A-modules
$\mathbf{Z}_\ell$, $\mathcal{O}_E$	systems (F_n) of constructible $\mathcal{O}_E/\pi_E^n$-sheaves with $F_{n+1} \otimes \mathcal{O}_E/\pi_E^n \simeq F_n$
$\mathbf{Q}_\ell$, E	the same systems up to isogeny, that is, with ℓ inverted
$\overline{\mathbf{Q}}_\ell$	a colimit of the categories for all finite $E/\mathbf{Q}_\ell$

On the pro-étale site, by contrast, one local condition covers all of them. Namely, a Λ_X-module is constructible if, on the strata of a finite stratification of X, it is pro-étale locally of the form $\Lambda_X \otimes_{\Lambda(*)} P$ for a finitely presented module P over $\Lambda(*)$, the ring of sections over the point (Definition 3.1). For finite A these are the classical constructible sheaves, while for E and $\overline{\mathbf{Q}}_\ell$ they are lisse on the strata. For $\mathcal{O}_E$, on the other hand, the modules P may have torsion, for example $P = \mathcal{O}_E/\pi_E$.

The six operations will act on $D^b_{\mathrm{ctf}}(X, \Lambda)$, bounded complexes with constructible cohomology and finite Tor amplitude. For Λ not a finite ring the Tor condition is automatic, that is, $D^b_{\mathrm{ctf}}(X, \Lambda) = D^b_c(X, \Lambda)$ (Proposition 3.12). For a finite ring, however, it matters: over $\mathbf{Z}/\ell^2$ the constructible sheaf $\mathbf{F}_\ell$ has infinite Tor dimension (Section 3.2 below).

Over $\mathcal{O}_E$ the passage to torsion is a criterion: for Noetherian X, a complex is constructible exactly when it is derived complete and its reduction modulo π_E is constructible (Proposition 3.11). The criterion is not formal, because the local models of the reductions modulo π_E^n must be made compatible in n (Proposition 3.10, Section 3.3). This description is due to Bhatt and Scholze [BS15].

Over E and $\overline{\mathbf{Q}}_\ell$, as the nodal curve shows, lattices exist only after stratifying, and this is enough. After descending from $\overline{\mathbf{Q}}_\ell$ to a finite extension E of $\mathbf{Q}_\ell$ and refining a stratification, a lisse sheaf has a lattice on each stratum (Lemma 3.24), and the reductions of the lattice modulo π_E^n are classical torsion sheaves (Lemma 3.25).

On a connected normal scheme no stratification is needed: there a lattice exists globally, and lisse E-sheaves are continuous representations of $\pi_1(X, \bar{x})$ (Proposition 13.10*, proved in Part III).

Dévissage along strata also shows that Hom out of a constructible complex commutes with filtered colimits of complexes bounded below by a common degree (Lemma 3.22); this continuity is used, together with descent to a finite E, in Chapter 5*.

Throughout this chapter schemes are again separated and of finite type over k, as in our conventions.

3.1 Derived functors

Four of the six operations, f^*, Rf_*, $\otimes^{\mathbf{L}}$ and internal Hom, exist for every condensed ring Λ on the whole unbounded derived categories by the usual resolutions. The remaining two need more work: $Rf_!$ is built from proper direct image and extension by zero (Section 5.4*), and its right adjoint comes from Brown representability.

Sheaves of Λ_X-modules can be viewed as module objects over the ring object Λ_X in $\mathrm{Sh}(X_{\mathrm{pro\acute{e}t}}, \mathrm{Ab})$, with the tensor product over Λ_X given by the coequalizer of the two actions $M \otimes \Lambda_X \otimes N \rightrightarrows M \otimes N$. The category of Λ_X-modules is Grothendieck abelian. Hence complexes have K-injective resolutions, which define Rf_* and $R\mathcal{H}om$ [Stacks, 07A5, 08J7]. They also have K-flat resolutions, which define $\otimes^{\mathbf{L}}$, and $K \otimes^{\mathbf{L}} -$ is left adjoint to $R\mathcal{H}om(K, -)$ [Stacks, 06YL].

The pullback of a Λ_Y-module M along $f : X \to Y$ is $f^*M = \Lambda_X \otimes_{f^{-1}\Lambda_Y} f^{-1}M$. Since $f^{-1}\Lambda_Y \simeq \Lambda_X$ (Lemma 2.54*), this is just $f^{-1}M$. Therefore $f^* : \mathrm{Mod}(Y_{\mathrm{pro\acute{e}t}}, \Lambda) \to \mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$

is exact, and on derived categories it is left adjoint to $Rf_* : D(X_{\text{pro\'et}}, \Lambda) \to D(Y_{\text{pro\'et}}, \Lambda)$. For an open immersion $j : U \to X$, extension by zero $j_! : \mathrm{Mod}(U_{\text{pro\'et}}, \Lambda) \to \mathrm{Mod}(X_{\text{pro\'et}}, \Lambda)$ is exact and left adjoint to j^*.

Constructible sheaves will be built from lisse sheaves on the strata of a stratification, each extended by zero to all of X (Lemma 3.18 and Corollary 3.20). We fix this extension by zero now. Let $a : S \hookrightarrow X$ be a locally closed immersion, for instance the inclusion of a stratum. Then S is open in its closure, so a factors as $a = i \circ j$, with $j : S \hookrightarrow \overline{S}$ the open immersion into the closure and $i : \overline{S} \hookrightarrow X$ the closed immersion. We put

$$a_! = i_* j_! : \mathrm{Mod}(S_{\text{pro\'et}}, \Lambda) \longrightarrow \mathrm{Mod}(X_{\text{pro\'et}}, \Lambda).$$

For a Λ_S-module L, the sheaf $a_!L$ restricts to L on S, and its restrictions to $\overline{S} \setminus S$ and to $X \setminus \overline{S}$ are zero. Indeed $a^* a_! = j^* i^* i_* j_! = \mathrm{id}$, the sheaf $j_!L$ vanishes on $\overline{S} \setminus S$, and i_* of a sheaf vanishes on $X \setminus \overline{S}$ (Proposition 2.58*, which applies because the open immersions involved are quasi-compact, X being Noetherian). The same proposition shows that $j_!$ and i_* are exact, so $a_!$ is exact. It therefore acts on complexes term by term and defines $a_! : D(S_{\text{pro\'et}}, \Lambda) \to D(X_{\text{pro\'et}}, \Lambda)$ with no derived functor needed. Once $Rf_!$ is constructed (Section 5.4*), this functor is $Ra_!$.

3.2 Constructibility

With finite coefficients a constructible sheaf is, on the strata of a finite stratification, locally constant with finite stalks. The six operations, however, will act on constructible complexes of finite Tor amplitude for every coefficient ring, so we need a definition of constructibility that covers all coefficient rings at once. We give it by a local condition (Definition 3.1). This condition has two parts, a stratification and a local model that replaces local constancy, and the complexes we use must in addition have finite Tor amplitude. As we explain below, each of the three conditions answers a specific need. For $\mathcal{O}_E$ we then show that constructibility can be tested on the reduction modulo π_E (Proposition 3.11). For a morphism $g : Z \to X$ and a Λ_X-module M we write $M|_Z = g^*M$, a Λ_Z-module.

A stratification cannot be avoided, already for sheaves coming from geometry: for $j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1$, the sheaf $j_!\Lambda$ is zero at the origin and Λ elsewhere, so it is not locally constant on $\mathbf{A}^1$.

Local constancy itself has to be rephrased. For instance, for $\Lambda = \mathbf{Z}_\ell$, "locally constant" with value a $\mathbf{Z}_\ell$-module would mean locally a constant sheaf with value the discrete group, but on $\operatorname{Spec} \mathbf{F}_q$ such a sheaf has $H^1 = 0$ (Section 1.2*), whereas the answer we want is $\mathbf{Z}_\ell$. We therefore ask instead that M be locally of the form $\Lambda_U \otimes_{\Lambda(*)} P$, with P a finitely presented module over the ring $\Lambda(*)$ of sections over the point, as in the opening of this chapter. This keeps the ℓ-adic topology. For finite Λ it is local constancy, and for E it says that M is locally free of finite rank. In this formulation, finite presentation of P replaces finiteness of the stalks.

Finally, the six operations should commute with change of coefficients, and this needs finite Tor amplitude. Over $\mathbf{Z}/\ell^2$ the sheaf $\mathbf{F}_\ell$ is constructible, but $\mathbf{F}_\ell \otimes^{\mathbf{L}}_{\mathbf{Z}/\ell^2} \mathbf{F}_\ell$ has cohomology $\mathbf{F}_\ell$ in every degree ≤ 0. With the Tor condition imposed, by contrast, the category D^b_{ctf} is stable under change of coefficients and under $\otimes^{\mathbf{L}}$, and for $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$ the Tor condition is automatic (Proposition 3.12).

Definition 3.1 (Constructible sheaves)**.** Let Λ be a commutative condensed ring.

1. Consider the following condition on a Λ_X-module M: there is a pro-étale cover $\{U_i \to X\}$ and finitely presented modules P_i over the ring $\Lambda(*)$ of sections over the point such that

$$M|_{U_i} \simeq \Lambda_{U_i} \otimes_{\Lambda(*)} P_i.$$

Here $\Lambda(*)$ and P_i are constant sheaves on U_i, acting through $\Lambda(*) \to \Lambda_{U_i}$. Equivalently, on each U_i the module M has a presentation by a finite matrix with entries in $\Lambda(*)$.

2. For qcqs X, a Λ_X-module M is *constructible* if there is a finite stratification of X by constructible locally closed subschemes such that on each stratum S the restriction $M|_S$ satisfies the condition of (1). On an arbitrary scheme this condition is imposed locally on affine opens.

We write $\mathrm{Cons}(X, \Lambda) \subset \mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ for the full subcategory of constructible sheaves, and $D_c^b(X, \Lambda) \subset D(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ for the full subcategory of bounded complexes with constructible cohomology sheaves. A complex K has *Tor amplitude in* $[a, b]$ if $K \otimes^{\mathbf{L}}_{\Lambda_X} N$ has cohomology only in $[a, b]$ for every sheaf N of Λ_X-modules. We write $D^b_{\mathrm{ctf}}(X, \Lambda)$ for the full subcategory of $D_c^b(X, \Lambda)$ of complexes of finite Tor amplitude.

The modules that satisfy the condition of (1) on all of X are the analogue of local systems, and they are the building blocks on the strata.

Definition 3.2 (Lisse sheaves)**.** Let Λ be a commutative condensed ring. A Λ_X-module M is *lisse* if it satisfies the condition of Definition definition-constructible-sheaves on X itself, without a stratification. A complex $K \in D(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ is *lisse* if its cohomology sheaves are lisse.

For the individual coefficient rings the definitions take a familiar form.

Remark 3.3 (Constructible sheaves for the individual coefficient rings)**.**

1. For Noetherian X, every locally closed subset is constructible, so in Definition definition-constructible-sheaves any finite stratification by locally closed subschemes may be used.
2. For finite A and qcqs X, $D^b_{\mathrm{ctf}}(X, A)$ is the pullback of the usual étale category $D^b_{\mathrm{ctf}}(X_{\acute{e}t}, A)$ by Corollary 2.44*.
3. If $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$, the ring $\Lambda(*)$ is a field, so each P_i is a finite-dimensional vector space, and a lisse module is locally free of finite rank. In other words, a constructible E- or $\overline{\mathbf{Q}}_\ell$-sheaf is one that is lisse on the strata of a finite stratification [BS15, Definition 6.8.6].
4. If $\Lambda = \mathcal{O}_E$, locally on a stratum a constructible $\mathcal{O}_{E,X}$-module is $P \otimes_{\mathcal{O}_E} \mathcal{O}_{E,X}$ for a finitely generated $\mathcal{O}_E$-module P. For example, $\mathcal{O}_{E,X}/\pi_E$ is constructible.
5. For the admissible coefficients, the constructible sheaves form an abelian category. For finite A this is the classical theory of sheaves with finite stalks, and the other cases are Lemma 3.18, Corollary 3.20 and Lemma 5.19*, which we prove in Subsection 5.4.2*.
6. For finite A, a lisse A_X-module is the same as a locally constant sheaf with finite stalks. For $\Lambda = \mathcal{O}_E$ a lisse module may have torsion, such as $\mathcal{O}_{E,X}/\pi_E$; the lisse $\mathcal{O}_{E,X}$-modules without torsion are the locally free ones of finite rank, since a finitely generated torsion free $\mathcal{O}_E$-module is free.

Part (5) is special to the admissible coefficients. For a general condensed ring, kernels of maps between constructible sheaves can fail to be constructible, and then $D_c^b(X, \Lambda)$ is not triangulated.

3.3 Integral constructible complexes

Constructibility over $\mathcal{O}_E$ is defined by a local condition, but in practice we know a complex $K \in D(X_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E)$ through its reductions $K_n = K \otimes^{\mathbf{L}}_{\mathcal{O}_{E,X}} (\mathcal{O}_E/\pi_E^n)_X$ of Definition 2.61*. The criterion below says that K is constructible exactly when it is derived complete and K_1 is constructible. Write $\mathbf{F}_E = \mathcal{O}_E/\pi_E$ for the residue field, so that $K_1 \in D(X_{\mathrm{pro\acute{e}t}}, \mathbf{F}_E)$.

To prove the criterion we must produce, on each stratum, one complex of finite free $\mathcal{O}_E$-modules that is locally a model for all the K_n at once. Each K_n separately is easy to control: its bounds and its stratification are inherited from K_1 by dévissage along the powers of π_E, and on a stratum it is locally isomorphic to a constant perfect complex. The difficulty is that these local isomorphisms are chosen independently at each level, so they have to be made compatible before we pass to the limit. On a weakly contractible cover we do this by a Mittag–Leffler argument, which needs a uniform rate. Such a rate exists because a bounded complex of finite free modules over the discrete valuation ring $\mathcal{O}_E$ has finitely generated cohomology, and the torsion of this cohomology is killed by a fixed power of π_E. Throughout, we use the étale site only at the finite levels, through Theorem 2.43*.

Example 3.4 (The criterion over a point)**.** On the spectrum of an algebraically closed field, take for K the sheaf attached to an $\mathcal{O}_E$-module. The computation is then one with modules, and it shows where the uniform rate comes from. Resolving $\mathcal{O}_E/\pi_E^e$ by $[\mathcal{O}_E \xrightarrow{\pi_E^e} \mathcal{O}_E]$ in degrees $-1, 0$ and reducing modulo π_E, we find

$$\mathcal{O}_E \otimes^{\mathbf{L}}_{\mathcal{O}_E} \mathbf{F}_E = \mathbf{F}_E, \qquad \mathcal{O}_E/\pi_E^e \otimes^{\mathbf{L}}_{\mathcal{O}_E} \mathbf{F}_E = [\mathbf{F}_E \xrightarrow{0} \mathbf{F}_E] = \mathbf{F}_E \oplus \mathbf{F}_E[1] \quad (e \geq 1),$$

since π_E^e becomes zero modulo π_E. For $P = \mathcal{O}_E^r \oplus \bigoplus_{i=1}^s \mathcal{O}_E/\pi_E^{e_i}$ the reduction K_1 has $H^0 = \mathbf{F}_E^{r+s}$ and $H^{-1} = \mathbf{F}_E^s$. Thus $H^{-1}(K_1)$ counts the cyclic torsion summands. For $K = \bigoplus_{n\geq 1} \mathcal{O}_E/\pi_E^n$ we get $K_1 = \bigoplus_{n\geq 1}(\mathbf{F}_E \oplus \mathbf{F}_E[1])$. This is not constructible, because its cohomology is an infinite-dimensional vector space in degrees 0 and -1. In other words, a constructible K_1 allows only finitely many torsion summands. For P the single power $\pi_E^{\max e_i}$ then kills all the torsion, and this is the uniform rate.

The completeness hypothesis cannot be dropped either. The module E is π_E-divisible, so $E \otimes^{\mathbf{L}} \mathcal{O}_E/\pi_E^n = 0$ for all n. Hence $K_1 = 0$ is constructible, yet E is not a finitely generated $\mathcal{O}_E$-module. The criterion does not apply because E is not derived complete: its derived completion $R\varprojlim_n (E \otimes^{\mathbf{L}} \mathcal{O}_E/\pi_E^n)$ is zero.

Lemma 3.5 (Bounds for the finite reductions)**.** *Let X be Noetherian and let $K \in D(X_{\text{proét}}, \mathcal{O}_E)$ be derived complete with $K_1 \in D^b_c(X, \mathbf{F}_E)$. Choose $[a, b]$ containing the cohomology of K_1. Then every K_n lies in $D^b_c(X, \mathcal{O}_E/\pi_E^n)$ and has Tor amplitude in $[a, b]$, and K has cohomology only in $[a, b]$. If the cohomology sheaves of K_1 are locally constant on a stratum, then so are those of every K_n.*

Proof. Everything about K_n is obtained from K_1 by induction on n, using that K_n is an extension of K_{n-1} by K_1. The Tor bound needs an extra argument with minimal resolutions, and the bound on K itself then follows from the Milnor sequence.

The filtration of $\mathcal{O}_E/\pi_E^n$ by the powers of π_E has graded pieces $\mathbf{F}_E$. Tensoring its short exact sequences with K over $\mathcal{O}_{E,X}$, we get triangles $K_1 \to K_n \to K_{n-1}$, up to the identification $\pi_E^{n-1}\mathcal{O}_E/\pi_E^n \simeq \mathbf{F}_E$. Moreover, complexes with constructible cohomology in $[a, b]$ are closed under extensions in triangles, by the long exact cohomology sequence and the fact that constructible sheaves form a weak Serre subcategory. Induction on n now shows that K_n is bounded in $[a, b]$ with constructible cohomology. Since locally constant finite sheaves of modules also form a weak Serre subcategory, the same induction works on a fixed stratum, and a single stratification works for all n. By the comparison theorem for constructible sheaves (Theorem 2.43*), these finite sheaves and complexes are classical, that is, pulled back from the étale site.

For the Tor bound we use $K_n \otimes^{\mathbf{L}}_{\mathcal{O}_E/\pi_E^n} \mathbf{F}_E \simeq K_1$. At an étale geometric point $\bar{x}$, take a bounded above resolution of the stalk of K_n by finite free modules over the Artinian local ring $\mathcal{O}_E/\pi_E^n$.

Cancelling a two-term identity summand whenever a differential matrix has a unit entry, we obtain a minimal resolution, whose differentials have entries in (π_E), so they vanish after tensoring with $\mathbf{F}_E$. Since stalks commute with derived tensor products, the minimal resolution tensored with $\mathbf{F}_E$ computes the stalk of K_1 at $\bar{x}$, with zero differential. Its rank in degree q is therefore $\dim_{\mathbf{F}_E} H^q(K_1)_{\bar{x}}$. This vanishes outside $[a, b]$, and we conclude that the stalk has a bounded finite free resolution in $[a, b]$. Such a bound on stalks gives the same Tor bound on the étale site, and Corollary 2.44* carries it to the pro-étale site, for Tor against arbitrary sheaves of modules there.

By completeness $K = R\varprojlim K_n$, so the Milnor sequence (2.2) gives the lower bound and the vanishing above $b+1$. For $H^{b+1}(K)$ we use the top degree, where $H^b(K_n) = H^b(K_{n+1}) \otimes_{\mathcal{O}_E/\pi_E^{n+1}} \mathcal{O}_E/\pi_E^n$, so the transition maps are epimorphisms. Their $R^1\varprojlim$ vanishes by the last assertion of Proposition 2.24*, and this means that $H^{b+1}(K) = 0$. ∎

To make the local trivializations of the K_n compatible, we need two facts about complexes over $\mathcal{O}_E$: a compatible system of perfect complexes over the rings $\mathcal{O}_E/\pi_E^n$ comes from a single bounded finite free complex, and the images of the cohomology at finite levels stabilize at a uniform rate.

Lemma 3.6 (Perfect complexes over $\mathcal{O}_E$)**.** *For $n \geq 1$ let $M_n \in D(\mathcal{O}_E/\pi_E^n)$ be perfect, with maps $M_{n+1} \to M_n$ inducing isomorphisms $M_{n+1} \otimes^{\mathbf{L}}_{\mathcal{O}_E/\pi_E^{n+1}} \mathcal{O}_E/\pi_E^n \simeq M_n$. Then there is a bounded complex $P^\bullet$ of finite free $\mathcal{O}_E$-modules with isomorphisms $P^\bullet/\pi_E^n \simeq M_n$ in $D(\mathcal{O}_E/\pi_E^n)$, compatible with the transition maps.*

Proof. We use that a minimal complex over the local ring $\mathcal{O}_E/\pi_E^n$ is determined up to isomorphism of complexes, not only up to homotopy, so that the transition maps can be made into equalities of matrices and passed to the limit.

Represent each M_n by a minimal finite free complex, obtained by cancelling unit entries in the differential matrices. Its reduction modulo π_E has zero differential, so its rank in degree q is $\dim_{\mathbf{F}_E} H^q(M_1)$, which is independent of n and nonzero only in finitely many degrees. The given isomorphism $M_{n+1}/\pi_E^n \simeq M_n$ between minimal complexes is represented by a chain homotopy equivalence f, with homotopy inverse g. Since both complexes have zero differential modulo π_E, the relations $gf - 1 = dh + hd$ and $fg - 1 = dh' + h'd$ show that f is an isomorphism modulo π_E in each degree. By Nakayama's lemma its matrices are then invertible over $\mathcal{O}_E/\pi_E^n$. Lifting them to invertible matrices over $\mathcal{O}_E/\pi_E^{n+1}$ and changing bases, we may assume that the differential at level $n+1$ reduces exactly to the differential at level n, with the given transition in the derived category. Induction on n gives compatible matrices, and their limits define $P^\bullet$. Its differential squares to zero because it does so modulo every π_E^n, and the identifications with the M_n are compatible by construction. ∎

The second fact is used in the proof of Lemma 3.9 for the internal Hom complex $(P^\bullet)^\vee \otimes P^\bullet$, whose H^0 at level n consists of the maps $P^\bullet/\pi_E^n \to P^\bullet/\pi_E^n$ in the derived category. There it says that a map at level n which lifts far enough lifts to all levels, and how far is enough depends only on the torsion of the cohomology.

Lemma 3.7 (Images of cohomology at finite levels)**.** *Let $Q^\bullet$ be a bounded complex of finite free $\mathcal{O}_E$-modules, and let π_E^e kill the torsion of $H^{q+1}(Q^\bullet)$. Then for $m \geq n + e$*

$$\operatorname{im}\big(H^q(Q^\bullet/\pi_E^m) \to H^q(Q^\bullet/\pi_E^n)\big) = \operatorname{im}\big(H^q(Q^\bullet) \to H^q(Q^\bullet/\pi_E^n)\big).$$

Such an e exists, since $\mathcal{O}_E$ is a Noetherian discrete valuation ring and $H^{q+1}(Q^\bullet)$ is finitely generated.

Proof of Lemma 3.7. The two-term resolution of $\mathcal{O}_E/\pi_E^n$ gives an exact sequence

$$0 \to H^q(Q^\bullet)/\pi_E^n \to H^q(Q^\bullet/\pi_E^n) \to H^{q+1}(Q^\bullet)[\pi_E^n] \to 0.$$

For the transition from level m to level n, the induced map on the right-hand terms is multiplication by π_E^{m-n}, because the corresponding map of resolutions is the identity in degree zero and multiplication by π_E^{m-n} in degree -1. If $m - n \geq e$, this map is zero, since $H^{q+1}(Q^\bullet)[\pi_E^m]$ consists of torsion elements and π_E^e kills them. The image of $H^q(Q^\bullet/\pi_E^m)$ therefore lies in the left-hand term $H^q(Q^\bullet)/\pi_E^n$ at level n, which is the image of $H^q(Q^\bullet)$. The reverse inclusion holds because the map from $H^q(Q^\bullet)$ factors through level m. ∎

The bound $m \geq n + e$ cannot be improved. To see this, take $Q^\bullet = [\mathcal{O}_E \xrightarrow{\pi_E^e} \mathcal{O}_E]$ in degrees $0, 1$, so that $H^0(Q^\bullet) = 0$ and $H^1(Q^\bullet) = \mathcal{O}_E/\pi_E^e$. For $m \geq e$ the group $H^0(Q^\bullet/\pi_E^m)$ is $\pi_E^{m-e}\mathcal{O}_E/\pi_E^m$, and its image at level n is generated by π_E^{m-e} modulo π_E^n. This image is zero exactly when $m \geq n + e$, in agreement with the image of $H^0(Q^\bullet) = 0$.

At a single finite level the local structure is a statement about the étale site: a perfect complex with locally constant cohomology is locally constant. On a connected scheme its local model is moreover the same at every point.

Lemma 3.8 (Locally constant perfect complexes)**.** *Let S be a connected Noetherian scheme, let A be a finite local ring, and let $L \in D^b(S_{ét}, A)$ have finite Tor amplitude and cohomology sheaves that are locally constant with finite stalks. Let $\bar{s}$ be a geometric point of S and let M be a bounded complex of finite free A-modules with $M \simeq L_{\bar{s}}$ in $D(A)$. Then L is étale locally on S isomorphic to the constant complex $\underline{M}$.*

Proof. We first show that a model of a stalk spreads out to a neighbourhood, and then use connectedness to see that the model is the same everywhere.

Every stalk $L_{\bar{t}}$ has finite Tor amplitude and finite cohomology, hence is perfect over the local ring A, that is, represented by a bounded complex $M_{\bar{t}}$ of finite free A-modules. An isomorphism $M_{\bar{t}} \simeq L_{\bar{t}}$ is then a section at $\bar{t}$ of the zeroth cohomology sheaf of $R\mathcal{H}om(\underline{M_{\bar{t}}}, L)$; here the stalks of this sheaf are computed stalkwise, since $\underline{M_{\bar{t}}}$ is a bounded complex of finite free modules. We extend the section to an étale neighbourhood of $\bar{t}$ and lift it there to a morphism $\underline{M_{\bar{t}}} \to L$. Its cone has bounded locally constant finite cohomology and zero stalk at $\bar{t}$. Hence its finitely many cohomology sheaves vanish after shrinking the neighbourhood, and L is isomorphic to $\underline{M_{\bar{t}}}$ there. In particular, the stalks of L at all geometric points over the image of this neighbourhood are isomorphic to $M_{\bar{t}}$ in $D(A)$.

For each isomorphism class c in $D(A)$, let $S_c \subset S$ be the set of points at whose geometric points the stalk of L lies in c. By the previous paragraph and openness of étale morphisms, each S_c is open. Moreover, the sets S_c are disjoint and cover S. Since S is connected, $S = S_c$ for the class c of M. Thus every point has an étale neighbourhood on which $L \simeq \underline{M}$; compare [Stacks, 09BI]. ∎

On a weakly contractible object the local isomorphisms of the finite levels become global, and the only remaining problem is to choose them compatibly. For this we use the uniform rate of Lemma 3.7.

Lemma 3.9 (Integral models on weakly contractible schemes)**.** *Let W be an affine weakly contractible scheme, let $K \in D(W_{\text{proét}}, \mathcal{O}_E)$ be derived complete, and let $P^\bullet$ be a bounded complex of finite free $\mathcal{O}_E$-modules. Suppose that for every $n \geq 1$ there is an isomorphism $(P^\bullet/\pi_E^n)_W \simeq K_n$ in $D(W_{\text{proét}}, \mathcal{O}_E/\pi_E^n)$. Then $K \simeq \mathcal{O}_{E,W} \otimes_{\mathcal{O}_E} P^\bullet$.*

Proof. The hypothesis gives an isomorphism $(P^\bullet/\pi_E^n)_W \simeq K_n$ at each level n, but these isomorphisms need not be compatible with the reduction maps, so they do not assemble to a map into $K = R\varprojlim_n K_n$. What we need is a compatible family of maps, that is, an element of the inverse limit of the Hom groups between the reductions. Such an element lifts to a map $\mathcal{O}_{E,W} \otimes_{\mathcal{O}_E} P^\bullet \to K$, because K is derived complete. To find one that is an isomorphism at level 1, we show that these Hom groups satisfy the Mittag–Leffler condition with a rate e independent of n. Derived Nakayama then shows that the lift is an isomorphism.

Consider the tower of complexes

$$R\mathcal{H}om_{(\mathcal{O}_E/\pi_E^n)_W}((P^\bullet/\pi_E^n)_W, K_n) \in D(W_{\text{pro\'et}}, \mathcal{O}_E/\pi_E^n),$$

with transition maps induced by the reduction maps $K_{n+1} \to K_n$, and the tower of groups $\mathrm{Hom}_{D(W_{\text{pro\'et}},\mathcal{O}_E/\pi_E^n)}((P^\bullet/\pi_E^n)_W, K_n)$ obtained by applying $H^0R\Gamma(W,-)$. Since $P^\bullet$ is finite free, the first tower is the tower of reductions of $(\mathcal{O}_{E,W} \otimes_{\mathcal{O}_E} (P^\bullet)^\vee) \otimes^{\mathbf{L}}_{\mathcal{O}_{E,W}} K$.

These Hom groups can be infinite, so the Mittag–Leffler condition has to come from a uniform bound. Let $Q^\bullet = (P^\bullet)^\vee \otimes_{\mathcal{O}_E} P^\bullet$ and choose e such that π_E^e kills the torsion of $H^1(Q^\bullet)$. Let $N \geq m \geq n+e$ and choose one isomorphism $(P^\bullet/\pi_E^N)_W \simeq K_N$. Its reductions to levels m and n identify the three internal Hom complexes at levels $N \to m \to n$ with the constant complexes $Q^\bullet/\pi_E^N \to Q^\bullet/\pi_E^m \to Q^\bullet/\pi_E^n$. Their zeroth cohomology sheaves are therefore the constant sheaves on the groups $H^0(Q^\bullet/\pi_E^N) \to H^0(Q^\bullet/\pi_E^m) \to H^0(Q^\bullet/\pi_E^n)$. By Lemma 3.7 the images from levels N and m at level n agree, as subsheaves. Since W is weakly contractible, taking sections over W is exact, and so it commutes with taking cohomology and images. Hence the images in the Hom group at level n from levels N and m coincide as well. Since e does not depend on n, m or N, we obtain the Mittag–Leffler condition: for every n the images at level n are constant from level $n+e$ on.

To lift, note that each Hom group contains an isomorphism, and its reduction to level 1 is again an isomorphism. The stable image at level 1 is the image from level $1+e$, so it contains an invertible class. By the Mittag–Leffler property this class lifts to a compatible family $(\varphi_n)_n$ in the inverse limit of the Hom groups. The Milnor sequence gives a surjection

$$\begin{aligned} H^0 R\varprojlim_n R\Gamma\big(W, R\mathcal{H}om_{(\mathcal{O}_E/\pi_E^n)_W}((P^\bullet/\pi_E^n)_W, K_n)\big) \\ \longrightarrow \varprojlim_n \mathrm{Hom}_{D(W_{\text{pro\'et}},\mathcal{O}_E/\pi_E^n)}\big((P^\bullet/\pi_E^n)_W, K_n\big), \end{aligned}$$

whose source is $\mathrm{Hom}_{D(W_{\text{pro\'et}},\mathcal{O}_E)}(\mathcal{O}_{E,W} \otimes_{\mathcal{O}_E} P^\bullet, K)$. This holds because derived sections commute with inverse limits, K is complete, and tensoring with the finite free dual of $P^\bullet$ commutes with inverse limits. The family (φ_n) therefore lifts to a morphism $\mathcal{O}_{E,W} \otimes_{\mathcal{O}_E} P^\bullet \to K$ whose first reduction is an isomorphism. Since both objects are complete, derived Nakayama (Lemma 2.62*) shows that this morphism is an isomorphism. Here completeness of the source follows termwise from $\mathcal{O}_{E,W} = \varprojlim (\mathcal{O}_E/\pi_E^n)_W$. Indeed, the transition maps are epimorphisms, so this limit agrees with the derived limit by Proposition 2.24*. ∎

Proposition 3.10 (Local structure of integral constructible complexes)**.** *Let X be Noetherian and let $K \in D(X_{\text{pro\'et}}, \mathcal{O}_E)$ be derived complete with $K_1 \in D^b_c(X, \mathbf{F}_E)$. Then there is a finite stratification of X such that on each stratum S, pro-étale locally,*

$$K|_S \simeq \mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} P^\bullet$$

for a bounded complex $P^\bullet$ of finite free $\mathcal{O}_E$-modules.

Proof. We first trivialize each K_n locally, separately for each n, and then make the trivializations compatible. To begin, choose a finite stratification on which all cohomology sheaves of K_1 are locally constant, and refine it by connected components; there are finitely many of these, since the strata are Noetherian.

For every stratum S the restriction $K|_S$ is again derived complete with constructible first reduction. Indeed, restriction to an open subscheme preserves limits, and so does restriction to a closed subscheme on the pro-étale site (Proposition 2.58*). Moreover, both restrictions are exact and commute with the two-term cones defining the reductions. From now on we fix a connected stratum S.

Local trivializations at each finite level. By Lemma 3.5, each $K_n|_S$ is classical and of finite Tor amplitude, with locally constant finite cohomology, so we may work on $S_{\acute{e}t}$. Fix a geometric point $\bar{s}$. Then the stalk complex at $\bar{s}$ has a minimal finite free model M_n in the common interval $[a, b]$. By Lemma 3.8 the complex $K_n|_S$ is étale locally isomorphic to the constant complex $(M_n)_S$.

On stalks at $\bar{s}$, the reduction maps $K_{n+1} \to K_n$ induce maps $M_{n+1} \to M_n$ inducing isomorphisms $M_{n+1} \otimes^{\mathbf{L}}_{\mathcal{O}_E/\pi_E^{n+1}} \mathcal{O}_E/\pi_E^n \simeq M_n$, because the étale stalk functor is exact and symmetric monoidal. By Lemma 3.6 there is a bounded finite free complex $P^\bullet$ with $P^\bullet/\pi_E^n \simeq M_n$. Back on $S_{\mathrm{proét}}$, we get for each n local isomorphisms $(P^\bullet/\pi_E^n)_S \simeq K_n|_S$ in $D(S_{\mathrm{proét}}, \mathcal{O}_E/\pi_E^n)$, chosen separately for each n.

The integral isomorphism. Choose an affine weakly contractible cover $W \to S$. For each n, the local isomorphism becomes an isomorphism $(P^\bullet/\pi_E^n)_W \simeq K_n|_W$ on all of W, since by weak contractibility a covering by trivializing charts splits. Moreover, the restriction $K|_W$ is derived complete, and its reductions are the $K_n|_W$. By Lemma 3.9 we conclude that $K|_W \simeq \mathcal{O}_{E,W} \otimes_{\mathcal{O}_E} P^\bullet$. ∎

The local structure is the hard half of the following criterion.

Proposition 3.11 (The integral constructibility criterion)**.** *Let X be Noetherian. The following conditions on $K \in D(X_{\mathrm{proét}}, \mathcal{O}_E)$ are equivalent:*

1. *$K \in D^b_c(X, \mathcal{O}_E)$;*
2. *K is derived complete and $K_1 \in D^b_c(X, \mathbf{F}_E)$;*
3. *there is a finite stratification of X such that on each stratum S, pro-étale locally, $K|_S \simeq \mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} P^\bullet$ for a bounded complex $P^\bullet$ of finite free $\mathcal{O}_E$-modules;*
4. *$K \in D^b_{\mathrm{ctf}}(X, \mathcal{O}_E)$.*

In the classical adic setting, this description of the integral constructible category is [BS15, Proposition 6.8.11] and [Stacks, 09C7].

Proof of Proposition 3.11. The substantial implication is (2)⇒(3), which is Proposition 3.10. The others hold because the complexes $\mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} P^\bullet$ are complete, have finite Tor amplitude, and have constructible reductions. These properties survive extension by zero from strata. We prove (1)⇒(2)⇒(3)⇒(1) and (3)⇔(4).

(1) implies (2). We first treat a single constructible sheaf, locally on a stratum, then extend by zero from the strata, and finally pass to complexes.

A finitely generated $\mathcal{O}_E$-module is a finite sum of copies of $\mathcal{O}_E$ and of modules $\mathcal{O}_E/\pi_E^r$, so its sheaf has a two-term finite free resolution. The coefficient sheaf $\mathcal{O}_{E,U_i}$ on a member U_i of the cover in Definition definition-constructible-sheaves is derived complete. Indeed, it is the limit of its finite quotients by Lemma 2.51*, and since the transition maps between these quotients are surjective,

this limit agrees with the derived limit by Proposition 2.24*. As completeness is preserved by finite sums and cones, the modules $\mathcal{O}_{E,U_i} \otimes_{\mathcal{O}_E} P_i$ of Definition definition-constructible-sheaves are complete, and their reductions are bounded locally constant finite complexes. We may check these assertions on a pro-étale cover, because restriction to an object is a localization of the topos, and it preserves limits and the cones defining the reductions.

For a constructible sheaf, choose a finite stratification as in the definition. Then for a stratum $a : S \hookrightarrow X$, factored as $a = i \circ j$ as in Section 3.1, the open immersion j is quasi-compact because X is Noetherian. Extension by zero $a_! = i_* j_!$ preserves completeness, because it is exact, commutes with the cones defining the reductions, and preserves limits. Here, for $j_!$ the last point is the assertion about limits in Proposition 2.58*, see [Stacks, 09BP], while for i_* it holds because i_* is exact and a right adjoint. Using the finite filtration by open and closed pieces, we see that the whole sheaf is complete with constructible finite reductions. Applying this to the finitely many cohomology sheaves of K and using the bounded Postnikov triangles, we obtain (2).

(2) implies (3). This is Proposition 3.10.

(3) implies (1). Each of the finitely many strata is quasi-compact, so the local isomorphisms of (3) can be taken on a finite covering by quasi-compact pro-étale objects, and their bounds give a common cohomological interval. Since the map from the constant sheaf with value the discrete ring $\mathcal{O}_E$ to $\mathcal{O}_{E,S}$ is flat [Stacks, 099L], the cohomology of $\mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} P^\bullet$ is $\mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} H^q(P^\bullet)$. Here $H^q(P^\bullet)$ is finitely generated over $\mathcal{O}_E$, so these sheaves satisfy the condition of Definition definition-constructible-sheaves. This proves (1).

(3) is equivalent to (4). The complexes $\mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} P^\bullet$ have finite Tor amplitude, and since the stratification is finite there is a common interval. Since the restrictions to the strata are exact and jointly conservative, they detect vanishing after tensoring with any sheaf of modules, and (3) implies (4). Conversely, (4) implies (1) by definition of D^b_{ctf}, and hence (3) by the implications already proved. ∎

Lemma 3.25 compares $D^b_{\mathrm{ctf}}(X, \mathcal{O}_E)$ with the classical category of constructible $\mathcal{O}_E$-complexes [Stacks, 09C1], which is defined with modules over the constant sheaf with value the discrete ring $\mathcal{O}_E$.

3.4 Finite Tor amplitude

For the integral and rational rings the finite Tor amplitude condition is automatic.

Proposition 3.12 (Constructible complexes have finite Tor amplitude for ℓ-adic coefficients)**.** *For $\Lambda = \mathbf{Z}_\ell, \mathcal{O}_E, \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell$, we have*

$$D^b_{\mathrm{ctf}}(X, \Lambda) = D^b_c(X, \Lambda).$$

Proof. For $\mathcal{O}_E$, and hence for $\mathbf{Z}_\ell$, this is the equivalence of (1) and (4) in the criterion above. For the fields we show that every constructible sheaf is flat, which bounds the Tor amplitude of a bounded complex by its cohomological amplitude.

For a field $\Lambda(*)$, a constructible sheaf M is, on each stratum and locally on a cover $\{U_i\}$, of the form $\Lambda_{U_i} \otimes_{\Lambda(*)} P_i$ with P_i a finite-dimensional vector space, so it is locally a finite free Λ_{U_i}-module. Restriction to the strata is exact and the restrictions are jointly conservative, as in the proof of the criterion. Hence every constructible sheaf has Tor amplitude in $[0, 0]$. Finally, a bounded complex whose finitely many cohomology sheaves have finite Tor amplitude has finite Tor amplitude, by the triangles of its canonical filtration. ∎

For a finite ring that is not regular the condition is needed, as the $\mathbf{Z}/\ell^2$-module $\mathbf{F}_\ell$ of Section 3.2 shows. Moreover, finite Tor amplitude is a condition on the whole complex, not on its cohomology sheaves. The category D^b_{ctf} is thick, but in general it is not stable under standard truncations. For example, for $A = \mathbf{Z}/\ell^2$ the complex $[A \xrightarrow{\ell} A]$ is perfect, whereas its nonzero cohomology modules are $\mathbf{F}_\ell$, which are not perfect. For this reason, the compactness proofs for finite A work with bounded flat constructible models, and standard truncations are taken in D^b_c. The models exist by the next lemma.

Lemma 3.13 (Flat models and generators)**.** *Every $C \in D^b_{\mathrm{ctf}}(X, A)$ on $X_{\mathrm{pro\acute{e}t}}$ is represented by a bounded complex of flat constructible A-sheaves. A flat constructible sheaf has a finite stratification on whose strata it is locally of the form $\underline{M}$ with M a finite projective A-module. Consequently, $D^b_{\mathrm{ctf}}(X, A)$ is generated, using shifts, finite extensions and direct summands, by sheaves $a_!L$, where $a : S \hookrightarrow X$ is locally closed and L is locally of the form $\underline{M}$ with M a finite projective A-module.*

Proof. The flat model exists on the étale site by a classical result, and we pull it back. The generators then come from two filtrations: the stupid filtration of the model, and the filtration of each term by strata.

By Corollary 2.44*, $C = \nu_X^* C_{\acute{e}t}$ with $C_{\acute{e}t} \in D^b_{\mathrm{ctf}}(X_{\acute{e}t}, A)$, where $\nu_X^* : D(X_{\acute{e}t}, A) \to D(X_{\mathrm{pro\acute{e}t}}, A)$ is the comparison inverse image. By [Stacks, 0F4M] the complex $C_{\acute{e}t}$ has a bounded flat model, and since ν_X^* is exact on sheaves, its pullback is a bounded flat constructible model on $X_{\mathrm{pro\acute{e}t}}$. We choose a common stratification on which its finitely many terms are locally constant, and their stalks are then finite projective, because a finite flat module over the Noetherian ring A is finite projective. If F is a flat constructible sheaf, then the exact sequences for the successive open and closed strata express F as a finite extension of the sheaves $a_!L$ attached to its strata. Similarly, the stupid filtration of the bounded flat complex expresses C as a finite extension of shifts of its *terms*, and shifts and finite extensions preserve finite Tor amplitude. The same argument proves the corresponding statement on $X_{\acute{e}t}$, which we use in the proof of Proposition 5.14* before passing to the pro-étale site. ∎

3.5 Field coefficients

Until the end of this section, $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$, so that $D^b_{\mathrm{ctf}} = D^b_c$ and we can argue with constructible cohomology sheaves and standard truncations. Recall (after Definition 3.2) that a Λ_X-module is constructible exactly when it is lisse, that is locally free of finite rank for the pro-étale topology, on each stratum of a finite stratification of X. By dévissage we reduce statements about $D^b_c(X, \Lambda)$ to the simplest constructible sheaves, namely lisse sheaves extended by zero from a stratum, the sheaves $a_!L$ of Section 3.1. This dévissage first splits a complex into its cohomology sheaves by truncation, and then filters each constructible sheaf by strata.

For this dévissage we need constructible sheaves to form an abelian category (Lemma 3.18). In the proof we reduce a morphism of lisse E-sheaves to a morphism of integral lattices, and for $\overline{\mathbf{Q}}_\ell$ we first descend the morphism to a finite E. To prepare for this, in the next two lemmas we construct the lattices locally and carry out the descent.

Definition 3.14 (Lattices)**.** Let $E/\mathbf{Q}_\ell$ be finite and let L be a lisse E_X-module. A *lattice* in L is a lisse $\mathcal{O}_{E,X}$-submodule $M \subset L$ with $M[1/\ell] = L$.

Lemma 3.15 (Lattices in lisse E-sheaves)**.** *Let $E/\mathbf{Q}_\ell$ be finite and let L be a lisse E_X-module. The sheaf on $X_{\mathrm{pro\acute{e}t}}$ of lattices in L is represented by a separated surjective étale X-scheme, with a universal lattice. In particular, L admits a lattice étale locally on X.*

Proof. Where L is free of rank r, a lattice is the same as a lattice in E^r, so the sheaf of lattices is locally the constant sheaf on the discrete set $\mathrm{GL}_r(E)/\mathrm{GL}_r(\mathcal{O}_E)$. The representing scheme is obtained by descending the corresponding disjoint unions of copies of the trivializing objects.

The rank of L is a locally constant function on $|X|$. Indeed, on a trivializing covering the subsets of constant rank are open, closed and saturated, and they descend by fpqc descent of open subsets. So X is the disjoint union of the open and closed subsets on which the rank is constant. Since the sheaf of lattices restricts to the sheaf of lattices on each of them, we may assume that L has constant rank r.

We now construct the representing scheme. On a pro-étale affine V where L is free of rank r, the sheaf of lattices is the constant sheaf on the set

$$\mathrm{GL}_r(E)/\mathrm{GL}_r(\mathcal{O}_E)$$

of lattices in E^r, which is discrete because $\mathrm{GL}_r(\mathcal{O}_E)$ is an open subgroup of $\mathrm{GL}_r(E)$. To glue these constant sheaves, recall from Lemma 2.51* that on an affine object, sections of E_X are continuous functions to E, with bounded denominators by quasi-compactness. A change of basis on $V \times_X V$ is therefore a continuous function to $\mathrm{GL}_r(E)$. Since the stabilizer of a lattice is open, the image of each fixed lattice under this function is locally constant on $V \times_X V$. Now consider the disjoint union of copies of $V \times_X V$ indexed by the lattices in E^r. On each component, the preimages of the target components are then open and closed and form a disjoint open cover, so the induced map from this disjoint union to itself is a morphism of schemes. In this way the cocycle of the transition matrices defines a descent datum of schemes for these disjoint unions of copies of V.

For affine $V = \operatorname{Spec} B$, the disjoint union of copies of V indexed by a set I is ind-quasi-affine, since its finite subunions are affine open subschemes and together they cover it. Since descent along fpqc coverings is effective for ind-quasi-affine morphisms [Stacks, 0APK], the descent datum above is effective. Taking a finite affine trivializing covering of each affine open of X and gluing, we therefore obtain a scheme representing the sheaf of lattices. This scheme is separated and étale over X by [Stacks, 02KU, 02VN], and it is surjective because every free module has a lattice. By descent of sheaves of modules, this scheme carries a universal lattice. ∎

For example, in rank one the lattices in E are the $\pi_E^m \mathcal{O}_E$ for $m \in \mathbf{Z}$, so the discrete set of lattices is $E^\times/\mathcal{O}_E^\times \simeq \mathbf{Z}$. Hence for $L = E_X$ the representing scheme is $\coprod_{m\in\mathbf{Z}} X$.

Example 3.16 (The scheme of lattices on the nodal curve)**.** Let $E = \mathbf{Q}_\ell$ and let L be the sheaf on the nodal curve X from the start of this chapter, obtained from E_Y on the chain of lines $Y \to X$ by letting the generator of $\mathbf{Z}$ act by ℓ. Over Y the lattices in $L|_Y = E_Y$ are the constant lattices $\ell^m \mathbf{Z}_\ell$, so over Y the scheme of lattices is $\coprod_{m\in\mathbf{Z}} Y$. The generator of $\mathbf{Z}$ moves the copy indexed by m to the copy indexed by $m+1$ while translating Y. Hence the scheme of lattices over X, which is the quotient by $\mathbf{Z}$, is a single copy of Y. It is surjective and étale over X, as the lemma says, but it has no section. Indeed, a section would trivialize the $\mathbf{Z}$-torsor $Y \to X$, whereas Y is connected and $\coprod_{\mathbf{Z}} X$ is not. This is the absence of a global lattice seen geometrically, and the universal lattice lives on Y, where L becomes constant.

With lattices we pass from E to $\mathcal{O}_E$. For $\overline{\mathbf{Q}}_\ell$ we first need to come down to a finite extension E, and on a quasi-compact scheme this only involves finitely many matrix entries.

Lemma 3.17 (Descent of lisse $\overline{\mathbf{Q}}$-sheaves)**.** *A finite collection of lisse $\overline{\mathbf{Q}}_{\ell,X}$-modules and morphisms between them is obtained by extension of scalars from lisse E_X-modules and their morphisms, for one finite extension $E/\mathbf{Q}_\ell$.*

Proof. The data are described by finitely many matrices with entries in $\overline{\mathbf{Q}}_{\ell,X}$ over finitely many quasi-compact objects, and each entry already lies in some E_X. We make this precise.

Since X is quasi-compact, we may choose a finite covering by quasi-compact pro-étale affines on which all the modules are free; such refinements exist by [Stacks, 0988]. Their pairwise intersections $U_i \times_X U_j$ are quasi-compact since X is quasi-separated, and quasi-separated since they are subschemes of the affine schemes $U_i \times U_j$. On a quasi-compact quasi-separated object, taking sections commutes with filtered colimits, because coverings admit finite quasi-compact refinements and the sheaf condition involves only finite limits [Stacks, 0739]. By Lemma 2.51* we have $\overline{\mathbf{Q}}_{\ell,X} = \varinjlim_E E_X$. Hence the finitely many transition matrices, their inverses, and the matrices of the given morphisms have entries in E_X for a single finite E. Moreover, their cocycle and compatibility identities already hold over E_X, since $E_X \to \overline{\mathbf{Q}}_{\ell,X}$ is injective. Descent of sheaves of modules now gives the required lisse modules and morphisms. ∎

With these two lemmas we can show that constructible sheaves with field coefficients form an abelian category, and obtain the filtration by extensions by zero of lisse sheaves that the dévissage uses.

Lemma 3.18 (Abelian category of constructible sheaves)**.** *Let $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$.*

1. $\mathrm{Cons}(X, \Lambda)$ *is closed under kernels, cokernels and extensions in* $\mathrm{Mod}(X_{\text{proét}}, \Lambda)$*. In particular it is an abelian category, and its inclusion into* $\mathrm{Mod}(X_{\text{proét}}, \Lambda)$ *is exact.*
2. $\mathrm{Cons}(X, \Lambda)$ *is essentially small.*
3. *Every $F \in \mathrm{Cons}(X, \Lambda)$ has a finite filtration $0 = F_0 \subset F_1 \subset \cdots \subset F_m = F$ whose subquotients F_k/F_{k-1} are of the form $a_! L$, with $a : S \hookrightarrow X$ a locally closed immersion and L a lisse Λ_S-module.*

If X is a curve, part (3) amounts to an exact sequence $0 \to j_! L \to F \to i_* i^* F \to 0$, where j is the inclusion of a dense open subset on which F is lisse, $L = j^* F$, and i is the inclusion of the finitely many closed points of the complement.

Proof of Lemma 3.18. Part (1) is the substantial one. All three closure properties are local on a common stratification, so it suffices to treat lisse sheaves on a single stratum, and there the only difficulty is the kernel of a morphism. Parts (2) and (3), by contrast, are formal.

(1) Given finitely many constructible sheaves, we refine their stratifications to a common one. Then all three closure properties can be checked on a single stratum S.

Extensions. Let $0 \to F' \to F \to F'' \to 0$ be exact with F' and F'' lisse on S. After passing to a cover, F' and F'' are free. Since $F \to F''$ is surjective, after a further cover a basis of F'' lifts to sections of F. Hence the sequence splits locally, and F is locally free of finite rank.

Kernels and cokernels. Locally, a morphism of lisse sheaves is a matrix of continuous functions to E, and a priori its kernel need not be locally free. Once we choose lattices, however, the morphism becomes a constant matrix on a suitable pro-étale cover, because its reductions are maps of finite locally constant sheaves. The kernel and cokernel of a constant matrix over E are constant of finite rank, and since being lisse is a local property for the pro-étale topology, this suffices.

First let $\Lambda = E$ and let $f : L \to L'$ be a morphism of lisse sheaves on S. By Lemma 3.15 we may replace S by a connected quasi-compact affine étale neighbourhood V on which L and L' contain lattices M and N. On a finite quasi-compact pro-étale covering trivializing both lattices, the matrix entries of f have a common denominator, so $h = \pi_E^m f$ maps M into N for some $m \geq 0$.

The reductions $M_n = M/\pi_E^n M$ and $N_n = N/\pi_E^n N$ are finite locally constant étale sheaves, and $M = \varprojlim M_n$, $N = \varprojlim N_n$ by Lemma 2.51*, checked where M and N are free. Fix a geometric point $\bar{v}$ of V. Then the completed fibres $M_{\bar{v}} = \varprojlim (M_n)_{\bar{v}}$ and $N_{\bar{v}} = \varprojlim (N_n)_{\bar{v}}$ are finite free $\mathcal{O}_E$-modules with continuous $\pi_1(V, \bar{v})$-actions, and the reductions of h give an equivariant map $h_{\bar{v}}$ between them. To make h constant, note that each M_n and N_n becomes constant on a connected finite étale cover of the connected scheme V, and we may choose these covers to form a tower. Their inverse limit $V' \to V$ is a connected pro-étale covering, so maps between constant sheaves on V' are constant. On V' the lattices M and N are identified with the constant sheaves attached to $M_{\bar{v}}$ and $N_{\bar{v}}$, and h with the constant matrix $h_{\bar{v}}$. Hence on V' the kernel and the cokernel of $f = \pi_E^{-m} h$ are the constant sheaves attached to the kernel and the cokernel of $h_{\bar{v}} \otimes_{\mathcal{O}_E} E$. In particular they are lisse.

For $\Lambda = \overline{\mathbf{Q}}_\ell$, we descend L, L' and f to a finite extension $E/\mathbf{Q}_\ell$ by Lemma 3.17. Since the functor $\overline{\mathbf{Q}}_{\ell,X} \otimes_{E_X} -$ is exact, it takes the kernel and cokernel over E to those of f. Finally, restriction to strata is exact, so the kernel and cokernel of a morphism of constructible sheaves on X are constructible.

(2) We choose the sites in a fixed universe. Up to isomorphism, a constructible sheaf is then determined by a finite stratification and, on each stratum, descent data of finite rank on a covering, and these data form a set.

(3) By Noetherian induction we refine the stratification of F so that there is an open stratum $j : U \hookrightarrow X$ with closed complement $i : Z \hookrightarrow X$ that is again a union of strata. The exact sequence $0 \to j_! j^* F \to F \to i_* i^* F \to 0$ separates the stratum U from the rest, and here $j^* F$ is lisse. Applying the same step to $i^* F$ on Z and iterating, we obtain a finite filtration of F whose subquotients are $a_!(F|_S)$ for the strata $a : S \hookrightarrow X$, and each $F|_S$ is lisse. ∎

Example 3.19 (A running example)**.** We use the same few schemes throughout:

$$\{0\} \xrightarrow{i} \mathbf{A}^1_k \xleftarrow{j} \mathbf{G}_{m,k}, \qquad \mathbf{A}^1_k \xrightarrow{j_\infty} \mathbf{P}^1_k.$$

The sheaf $j_! \Lambda_{\mathbf{G}_m}$ is constructible for the stratification $\mathbf{A}^1 = \mathbf{G}_m \sqcup \{0\}$. Its stalk is Λ at a geometric point of $\mathbf{G}_m$ and 0 at the origin, so it is not lisse on $\mathbf{A}^1$, and the sequence of part (3) is the localization sequence

$$0 \to j_! \Lambda_{\mathbf{G}_m} \to \Lambda_{\mathbf{A}^1} \to i_* \Lambda \to 0.$$

We return to this example for compact supports (after Lemma 6.4*), for $i^!$ (Example 6.25*) and for the trace (Section 9.6*).

Exercises. (1) Compute the stalks of $a_! L$ for a locally closed immersion a and a lisse L. (2) For $K = \Lambda_X[1]$, in which degree is the nonzero cohomology sheaf? (3) Show that $j_! \Lambda_{\mathbf{G}_m}$ has no nonzero global sections on $\mathbf{A}^1_k$.

For complexes we get the following. Since constructible sheaves form an abelian subcategory, complexes with constructible cohomology are stable under cones, and the filtration of part (3) together with the truncation triangles generates them from the sheaves $a_! L$.

Corollary 3.20 (Generators of D^b_c with field coefficients)**.** *Let $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$. Then $D^b_c(X, \Lambda)$ is a triangulated subcategory of $D(X_{\text{pro\'et}}, \Lambda)$, and every object of $D^b_c(X, \Lambda)$ is obtained by finitely many cones from objects $a_! L[n]$, where $a : S \hookrightarrow X$ is a locally closed immersion, L is a lisse Λ_S-module and $n \in \mathbf{Z}$.*

Proof. The first assertion is closure under cones, which we check on cohomology sheaves. Let $K \to K' \to K'' \to K[1]$ be a distinguished triangle with K, K' in $D_c^b(X, \Lambda)$. By the long exact cohomology sequence, each $\mathcal{H}^i(K'')$ is an extension of $\ker(\mathcal{H}^{i+1}K \to \mathcal{H}^{i+1}K')$ by $\operatorname{coker}(\mathcal{H}^i K \to \mathcal{H}^i K')$. Both are constructible by Lemma lemma-constructible-sheaves, and hence so is $\mathcal{H}^i(K'')$. Closure under shifts is clear, since shifts only renumber the cohomology sheaves.

For the second assertion, we argue by induction on the number of nonzero cohomology sheaves of K. If i is the largest degree with $\mathcal{H}^i(K) \neq 0$, the truncation triangle $\tau_{<i}K \to K \to \mathcal{H}^i(K)[-i] \to$ exhibits K as the cone of a map $\mathcal{H}^i(K)[-i-1] \to \tau_{<i}K$, and $\tau_{<i}K$ has fewer nonzero cohomology sheaves. This reduces the claim to a single shifted constructible sheaf, and for it the filtration from Lemma lemma-constructible-sheaves yields finitely many short exact sequences, that is, distinguished triangles, with terms $a_!L$. ∎

3.6 Continuity of Ext

A morphism of $\overline{\mathbf{Q}}_\ell$-complexes should be defined over some finite extension E of $\mathbf{Q}_\ell$. Since $\overline{\mathbf{Q}}_{\ell,X} = \varinjlim_E E_X$, this amounts to asking that Hom out of a constructible complex commute with a filtered colimit, where a filtered colimit of complexes means the termwise colimit of a filtered diagram of complexes of sheaves. We prove this continuity for complexes with a common lower bound. In this form it is the pro-étale analogue of the classical fact that cohomology of a quasi-compact quasi-separated scheme commutes with filtered colimits. Besides the descent to a finite coefficient field below, continuity is used in two places. The first is the uniform bound for Ext groups, which is proved first for integral lattices, so we need continuity for $\mathcal{O}_E$ as well. The second is the compactness argument for E and $\overline{\mathbf{Q}}_\ell$ (Section 5.2*).

For cohomology and higher direct images the statement is Corollary 2.29*, which holds for every sheaf of rings. To reach Hom, recall that a constructible sheaf has a finite filtration by sheaves $a_!L$ with L lisse on a stratum S (Lemma lemma-constructible-sheaves). Using sections with support in a closed subscheme, we turn Hom out of $a_!L$ into cohomology on S of a complex built from the target by Rj_* and restrictions, and the comparison then goes through a hypercohomology spectral sequence. Because of the common lower bound, only finitely many of its terms contribute to a given degree, and each of these terms commutes with the colimit.

Lemma 3.21 (Sections with closed support)**.** *Let Λ be admissible, let $i : Z \hookrightarrow X$ be a closed immersion with open complement $j : V \hookrightarrow X$, and let $R\underline{\Gamma}_Z : D(X_{\text{pro\'et}}, \Lambda) \to D(Z_{\text{pro\'et}}, \Lambda)$ be the local cohomology functor (Definition 2.59*),*

$$R\underline{\Gamma}_Z K = i^* \operatorname{fib}(K \longrightarrow Rj_*j^*K).$$

For a sheaf M of Λ_X-modules on $X_{\text{pro\'et}}$, let $\underline{\Gamma}_Z(M) = i^* \ker(M \to j_*j^*M)$, the sheaf on Z of sections of M with support in Z.

1. *$R\underline{\Gamma}_Z$ is the right derived functor of $\underline{\Gamma}_Z$. Consequently the adjunction of Lemma 2.60* holds in the form $R\mathrm{Hom}_X(i_*N, K) = R\mathrm{Hom}_Z(N, R\underline{\Gamma}_Z K)$.*
2. *Let $a = i \circ u : S \hookrightarrow X$ with $u : S \hookrightarrow Z$ an open immersion, and let L be a finite locally projective Λ_S-module with dual $L^\vee$. Then naturally in $K \in D(X_{\text{pro\'et}}, \Lambda)$*

$$R\mathrm{Hom}_X(a_!L, K) = R\Gamma\big(S, L^\vee \otimes u^* R\underline{\Gamma}_Z K\big). \tag{3.1}$$

Proof. Lemma 2.60* gives the adjunction between i_* and $R\underline{\Gamma}_Z$ on derived categories. What is new here is that $R\underline{\Gamma}_Z$ is the right derived functor of $\underline{\Gamma}_Z$, and this upgrades the adjunction to $R\mathrm{Hom}$ complexes and yields formula (3.1). For this we show two things: $\underline{\Gamma}_Z$ is right adjoint

to the exact functor i_* on sheaves, and $M \to j_*j^*M$ is surjective for injective M. The second means that on injective sheaves the kernel of this map is also its fibre.

We use that i^* and i_* are exact, that $i^*i_* = \mathrm{id}$, and that every sheaf M sits in the localization sequence $0 \to j_!j^*M \to M \to i_*i^*M \to 0$ [Stacks, 09AA, 09AH, 09BM]. The last fact shows that a sheaf whose restriction to V vanishes is of the form i_* of a sheaf on Z. In particular $\ker(M \to j_*j^*M) = i_*\underline{\Gamma}_Z(M)$.

Adjunction on sheaves. The functor $\underline{\Gamma}_Z$ is right adjoint to $i_* : \mathrm{Mod}(Z_{\mathrm{pro\acute{e}t}}, \Lambda) \to \mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$. Indeed, a map $i_*N \to M$ composes to zero with $M \to j_*j^*M$, because $j^*i_*N = 0$, so it factors through $\ker(M \to j_*j^*M) = i_*\underline{\Gamma}_Z(M)$. Since i_* is fully faithful, such factorizations are the maps $N \to \underline{\Gamma}_Z(M)$. Moreover, as i_* is exact, its right adjoint $\underline{\Gamma}_Z$ preserves K-injective complexes.

Surjectivity on injectives. For an injective sheaf I the unit $I \to j_*j^*I$ of the adjunction $j^* \dashv j_*$ on $\mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ is surjective, because it has a right inverse. To construct it, extend the inclusion $j_!j^*I \to I$ along $j_!j^*I \hookrightarrow j_*j^*I$ to a map $s : j_*j^*I \to I$, which is possible because I is injective. Then the composite $j_*j^*I \xrightarrow{s} I \to j_*j^*I$ restricts to the identity on V. Since j_* is fully faithful, an endomorphism of j_*j^*I is determined by its restriction to V, and so this composite is the identity.

Comparison with the fibre. The right derived functor of $\underline{\Gamma}_Z$ agrees with $i^*\,\mathrm{fib}(K \to Rj_*j^*K)$. To see this, represent K by a K-injective complex $I^\bullet$ with injective terms [Stacks, 079P]. Then $\underline{\Gamma}_Z(I^\bullet)$ computes the right derived functor of $\underline{\Gamma}_Z$ at K. On the other side, the complex $j^*I^\bullet$ is K-injective, because the left adjoint $j_!$ of j^* is exact. Hence $j_*j^*I^\bullet$ represents Rj_*j^*K. By the surjectivity just proved, the sequences

$$0 \to i_*\underline{\Gamma}_Z(I^n) \to I^n \to j_*j^*I^n \to 0$$

are exact termwise, and they define a distinguished triangle $i_*\underline{\Gamma}_Z(I^\bullet) \to K \to Rj_*j^*K \to i_*\underline{\Gamma}_Z(I^\bullet)[1]$. Applying i^* and using $i^*i_* = \mathrm{id}$ identifies $\underline{\Gamma}_Z(I^\bullet)$ with $i^*\,\mathrm{fib}(K \to Rj_*j^*K)$.

Part (1). With $I^\bullet$ as above, the complex $\underline{\Gamma}_Z(I^\bullet)$ is K-injective by the adjunction on sheaves, and it represents $R\underline{\Gamma}_ZK$ by the comparison with the fibre. The adjunction on complexes therefore gives

$$R\mathrm{Hom}_X(i_*N, K) = \mathrm{Hom}^\bullet(i_*N, I^\bullet) = \mathrm{Hom}^\bullet(N, \underline{\Gamma}_Z(I^\bullet)),$$

whose right-hand side is $R\mathrm{Hom}_Z(N, R\underline{\Gamma}_ZK)$.

Formula (3.1). Since $a_! = i_*u_!$, part (1), the adjunction $u_! \dashv u^*$ and the identity $R\mathcal{H}om(L, -) = L^\vee \otimes -$ for the finite locally projective sheaf L give $R\mathrm{Hom}_X(i_*u_!L, K) = R\mathrm{Hom}_Z(u_!L, R\underline{\Gamma}_ZK) = R\Gamma(S, L^\vee \otimes u^*R\underline{\Gamma}_ZK)$. ∎

Formula (3.1) expresses Hom out of $a_!L$ through cohomology on the stratum S and the functors i^*, u^* and Rj_*. Each of these commutes with filtered colimits of uniformly bounded below complexes, and from this we deduce the continuity of Ext.

Lemma 3.22 (Continuity of Ext)**.** *Let $(K_\alpha^\bullet)$ be a filtered diagram of complexes of Λ_X-modules that all vanish below a fixed degree, and let M be one of the following:*

1. *an object of $D_c^b(X, \Lambda)$, where $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$;*
2. *a finite extension of sheaves $a_!L_0$, with $a : S \hookrightarrow X$ locally closed and L_0 a finite locally free $\mathcal{O}_{E,S}$-module, where $\Lambda = \mathcal{O}_E$.*

Then for every integer n the natural map

$$\varinjlim_{\alpha} \mathrm{Hom}(M, K_\alpha^\bullet[n]) \longrightarrow \mathrm{Hom}(M, (\varinjlim_{\alpha} K_\alpha^\bullet)[n])$$

is an isomorphism.

Applied to sheaves, which are complexes in degree 0, the lemma says that for $M \in \mathrm{Cons}(X, \Lambda)$, $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$, the functors $\mathrm{Ext}^n(M, -)$ commute with filtered colimits of Λ_X-modules.

Proof of Lemma 3.22. We reduce to a single generator $a_!L$ and then use (3.1) to turn Hom into cohomology on the stratum S, where Corollary 2.29* applies. The common lower bound is used at the end, to compare the abutments of two spectral sequences.

In both cases M is obtained by finitely many extensions and shifts from sheaves $a_!L$, with $a : S \hookrightarrow X$ a locally closed immersion and L a finite locally free Λ_S-module: in case (1) by Corollary 3.20, and in case (2) by assumption. Both sides are cohomological functors of M. By the five lemma, the class of objects M for which the claim holds is therefore closed under extensions and shifts, so it suffices to treat $M = a_!L$, with $a = i \circ u : S \hookrightarrow Z \hookrightarrow X$ as in Lemma 3.21. The only property of L used below is that $L^\vee \otimes -$ is exact.

By (3.1), we must show that the cohomology of $R\Gamma(S, L^\vee \otimes u^* R\underline{\Gamma}_Z K)$ commutes with the colimit, where $K = \varinjlim_\alpha K_\alpha^\bullet$. To this end we apply the exact functor $L^\vee \otimes u^* i^*(-) : D(X_{\mathrm{pro\acute{e}t}}, \Lambda) \to D(S_{\mathrm{pro\acute{e}t}}, \Lambda)$ to the triangle $i_* R\underline{\Gamma}_Z K \to K \to Rj_* j^* K \to$ and to the corresponding triangles for the $K_\alpha^\bullet$. By the long exact sequence and the five lemma, it suffices to treat the two terms

$$R\Gamma\big(S, L^\vee \otimes u^* i^* K\big) \quad \text{and} \quad R\Gamma\big(S, L^\vee \otimes u^* i^* Rj_* j^* K\big).$$

For the first term, Corollary 2.29* on S applies, because the functor $L^\vee \otimes u^* i^*(-)$ is exact, commutes with colimits and preserves the lower bound. For the second term, consider the hypercohomology spectral sequence

$$E_2^{p,r} = H^p\big(S, L^\vee \otimes u^* i^* R^r j_*(j^* K)\big) \Longrightarrow H^{p+r}\big(S, L^\vee \otimes u^* i^* Rj_* j^* K\big)$$

which is functorial in K. The sheaves $R^r j_*(j^* K_\alpha)$ commute with the colimit by the second part of Corollary 2.29*, and $H^p(S, -)$ does by its first part. Hence, after passing to the colimit, the map of spectral sequences is an isomorphism on E_2. The abutments agree as well, since by the common lower bound only finitely many $E_2^{p,r}$ contribute to a given total degree. ∎

3.7 Descent to a finite extension

The lattice arguments below need the discrete valuation ring $\mathcal{O}_E$ with its uniformizer π_E, but the valuation ring of $\overline{\mathbf{Q}}_\ell$ is not of this kind. Since $\overline{\mathbf{Q}}_\ell$ is the filtered union of the finite extensions E of $\mathbf{Q}_\ell$, we show instead that every object and every morphism of $D_c^b(X, \overline{\mathbf{Q}}_\ell)$ is defined over some finite E.

For $C \in D(X_{\mathrm{pro\acute{e}t}}, E)$ and a finite extension E'/E, the extensions of scalars are the complexes $C_{E'} = E'_X \otimes_{E_X} C$ and $C_{\overline{\mathbf{Q}}_\ell} = \overline{\mathbf{Q}}_{\ell,X} \otimes_{E_X} C$. These tensor products need not be derived, since E'_X and $\overline{\mathbf{Q}}_{\ell,X}$ are flat over E_X.

The classical counterpart is the fact that a continuous representation of a profinite group on $\overline{\mathbf{Q}}_\ell^r$, with the ℓ-adic topology, takes values in $\mathrm{GL}_r(E)$ for some finite $E/\mathbf{Q}_\ell$. Similarly, a constructible complex, or a morphism between two of them, is described by finitely many matrices with entries

in $\overline{\mathbf{Q}}_{\ell,X}$, and on a quasi-compact scheme finitely many such entries already lie in a single E_X. We use the next lemma whenever a statement for $\overline{\mathbf{Q}}_\ell$ is reduced to one for a finite E, as in Lemma lemma-integral-models and in the uniform Ext bound (Lemma 5.6*).

Lemma 3.23 (Descent to a finite coefficient field)**.**

1. *For $C, D \in D_c^b(X, E)$ and every integer r, extension of scalars induces an isomorphism*
$$\varinjlim_{E' \supset E} \mathrm{Hom}_{D(X_{\text{pro\'et}}, E')}(C_{E'}, D_{E'}[r]) \xrightarrow{\sim} \mathrm{Hom}_{D(X_{\text{pro\'et}}, \overline{\mathbf{Q}}_\ell)}(C_{\overline{\mathbf{Q}}_\ell}, D_{\overline{\mathbf{Q}}_\ell}[r]),$$
the colimit running over finite extensions E' of E in $\overline{\mathbf{Q}}_\ell$.
2. *Every object of $D_c^b(X, \overline{\mathbf{Q}}_\ell)$ is isomorphic to $K_{\overline{\mathbf{Q}}_\ell}$ for some $K \in D_c^b(X, E)$ and some finite $E/\mathbf{Q}_\ell$.*

Proof. Morphisms descend because $\overline{\mathbf{Q}}_{\ell,X}$ is the filtered colimit of the E'_X and Hom out of a constructible complex commutes with this colimit (Lemma 3.22). Objects are built from lisse sheaves by finitely many extensions, each classified by a morphism, and so they descend as well.

(1) Extension of scalars $E'_X \otimes_{E_X} -$ is left adjoint to restriction of scalars $D(X_{\text{pro\'et}}, E') \to D(X_{\text{pro\'et}}, E)$. Hence $\mathrm{Hom}_{D(X_{\text{pro\'et}}, E')}(C_{E'}, D_{E'}[r]) = \mathrm{Hom}_{D(X_{\text{pro\'et}}, E)}(C, D_{E'}[r])$, and similarly for $\overline{\mathbf{Q}}_\ell$. Represent D by a bounded complex $D^\bullet$. Then the complexes $D^\bullet \otimes_{E_X} E'_X$ form a filtered system of complexes of E_X-modules, all vanishing below the lower bound of $D^\bullet$, and their colimit is $D^\bullet \otimes_{E_X} \overline{\mathbf{Q}}_{\ell,X}$, which represents $D_{\overline{\mathbf{Q}}_\ell}$. Applying Lemma 3.22 to $C \in D_c^b(X, E)$, we get
$$\varinjlim_{E'} \mathrm{Hom}_{D(X_{\text{pro\'et}}, E)}(C, D_{E'}[r]) \xrightarrow{\sim} \mathrm{Hom}_{D(X_{\text{pro\'et}}, E)}(C, D_{\overline{\mathbf{Q}}_\ell}[r]),$$
which is the claim after the adjunction.

(2) First let K be a constructible $\overline{\mathbf{Q}}_{\ell,X}$-module, with filtration $N_1 \subset \cdots \subset N_m = K$ as in Lemma lemma-constructible-sheaves. We descend N_k by induction on k. The subquotient $N_k/N_{k-1} = a_! L$ descends, because L does by Lemma 3.17 and $a_!$ commutes with extension of scalars. Suppose N_{k-1} and N_k/N_{k-1} descend to E. Then the extension $0 \to N_{k-1} \to N_k \to N_k/N_{k-1} \to 0$ gives a distinguished triangle, classified by a morphism $N_k/N_{k-1} \to N_{k-1}[1]$. By (1) this morphism descends after a finite extension of E. The fibre of the descended morphism, that is, its cone shifted by $[-1]$, is a model of N_k.

For a bounded complex K, we argue in the same way by induction on the number of nonzero cohomology sheaves. Let i be the largest degree with $\mathcal{H}^i(K) \neq 0$, and consider the truncation triangle $\tau_{<i}K \to K \to \mathcal{H}^i(K)[-i] \to$. Once $\tau_{<i}K$ and $\mathcal{H}^i(K)$ have models over a finite E, the connecting morphism $\mathcal{H}^i(K)[-i-1] \to \tau_{<i}K$ descends after a finite extension by (1), and the cone of the descended morphism is a model of K. Each step replaces E by a finite extension, so after finitely many steps K is defined over a single finite E. ∎

3.8 Lattices after stratification

We pass from statements with coefficients $\mathcal{O}_E/\pi_E^n$ to statements with coefficients E through integral lattices. A lisse E_X-module may fail to have a lattice on all of X, but it has one on each stratum of a finite stratification (Lemma 3.24), and this is enough for dévissage arguments. In particular, from the stratified statement we obtain the integral generators of Lemma lemma-integral-models. Lemma 3.24 is also the first step in the proofs of the uniform Ext bound and of the degree bound for proper direct image in Chapter 5*. On a connected normal scheme, by

contrast, there is a global lattice, and lisse E-sheaves are representations of $\pi_1(X, \bar{x})$; we prove this where it is used, in Proposition 13.10*.

On the nodal curve X of Example 3.16, the sheaf L has no global lattice, and one stratification step repairs this. On the open stratum $X \smallsetminus \{\text{node}\} \simeq \mathbf{G}_m$ the torsor is trivial, so L is constant there and has a lattice. The node, with its reduced structure, is the spectrum of a field, which is normal, and hence the restriction of L to the node has a lattice as well.

Lemma 3.24 (Lattices after a finite stratification)**.** *Let L be a lisse E_X-module. There is a finite stratification of X by locally closed subschemes S such that each $L|_S$ contains a lisse $\mathcal{O}_{E,S}$-submodule L_0 with $L_0[1/\ell] = L|_S$.*

In the proof we first find a lattice after a surjective étale base change and then push it down by the trace. For the first step we use that the lattices in E^r form a discrete set on which $\mathrm{GL}_r(E)$ acts continuously, which makes the sheaf of lattices in L an étale X-scheme. For the second step the base change must be finite étale, which we arrange by restricting to the strata of a finite stratification.

Proof of Lemma 3.24. Let $Y \to X$ be the étale X-scheme representing lattices in L (Lemma 3.15), with its universal lattice.

Pushing down by the trace. Choose finitely many quasi-compact affine opens of this scheme whose images cover X, and let $p : V \to X$ be their disjoint union. Then p is quasi-compact, separated, surjective and étale, and the universal lattice gives a lattice $M_0 \subset p^*L$. By [Stacks, 07RY, 07RZ], X has a finite stratification by locally closed subschemes over each of which p is finite étale and surjective. On such a stratum S, let $\mathrm{Tr} : p_*p^*L \to L$ be the trace of the finite étale map p over S, and define $L_0 \subset L|_S$ as the image of the composite

$$p_*M_0 \longrightarrow p_*p^*L \xrightarrow{\mathrm{Tr}} L.$$

To see that L_0 is a lattice, we split the finite étale cover locally on S into d copies of S. Then p_*M_0 is a direct sum of d lattices $M_1, \dots, M_d$ in L, and the trace is their sum, so $L_0 = M_1 + \cdots + M_d$. If we further trivialize L pro-étale locally, each M_k becomes a locally constant function to the discrete set of lattices in E^r, and after a common refinement each M_k is the constant sheaf attached to a lattice $N_k \subset E^r$. Then L_0 is the constant sheaf attached to $N_1 + \cdots + N_d$, which is a finite free $\mathcal{O}_E$-lattice spanning E^r. Hence L_0 is lisse and $L_0[1/\ell] = L|_S$. ∎

3.9 Integral models

To deduce statements over E and $\overline{\mathbf{Q}}_\ell$ from statements over $\mathcal{O}_E$, we need integral objects that generate $D^b_c(X, E)$ and $D^b_c(X, \overline{\mathbf{Q}}_\ell)$. We also want to know that our integral constructible complexes are those of the Stacks Project, so that results proved there apply, and we prove both in the next lemma. Recall from Proposition 3.11 in Section 3.3 that an object lies in $D^b_c(X, \mathcal{O}_E) = D^b_{\mathrm{ctf}}(X, \mathcal{O}_E)$ precisely when it is derived complete and its reduction modulo π_E is a bounded constructible complex pulled back from $X_{ét}$. The Stacks Project works instead with the category $D_{\mathrm{cons}}(X, \mathcal{O}_E)$ of [Stacks, 09C1], built from modules over the constant sheaf $\mathcal{O}_E$ rather than over $\mathcal{O}_{E,X}$.

Lemma 3.25 (Integral models)**.**

1. *Restriction of scalars along the map from the constant sheaf $\mathcal{O}_E$ to $\mathcal{O}_{E,X}$ is an equivalence between the derived complete objects of $D(X_{\mathrm{proét}}, \mathcal{O}_E)$ and the derived complete complexes of modules over the constant sheaf $\mathcal{O}_E$.*

2. *An object K of $D(X_{\text{pro\'et}}, \mathcal{O}_E)$ lies in $D^b_c(X, \mathcal{O}_E)$ if and only if its restriction of scalars lies in $D_{\text{cons}}(X, \mathcal{O}_E)$.*
3. *$D^b_c(-, \mathcal{O}_E)$ is preserved by $f^* : D(Y_{\text{pro\'et}}, \mathcal{O}_E) \to D(X_{\text{pro\'et}}, \mathcal{O}_E)$ for every morphism $f : X \to Y$, by $\otimes^{\mathbf{L}}_{\mathcal{O}_{E,X}}$ on $D(X_{\text{pro\'et}}, \mathcal{O}_E)$, and by $a_! : D(S_{\text{pro\'et}}, \mathcal{O}_E) \to D(X_{\text{pro\'et}}, \mathcal{O}_E)$ for every locally closed immersion $a : S \hookrightarrow X$.*
4. *Every object of $D^b_c(X, E)$ is obtained by finitely many cones and shifts from objects $(a_!L_0)[1/\ell]$, where $a : S \hookrightarrow X$ is a locally closed immersion and L_0 is a lisse $\mathcal{O}_{E,S}$-module that is locally free of finite rank. Every object of $D^b_c(X, \overline{\mathbf{Q}}_\ell)$ is obtained in the same way from the objects $\overline{\mathbf{Q}}_{\ell,X} \otimes_{\mathcal{O}_{E,X}} a_!L_0$, for some finite $E/\mathbf{Q}_\ell$.*

In (4) each L_0 satisfies the condition of Definition definition-constructible-sheaves, so it lies in $\text{Cons}(S, \mathcal{O}_E)$. By (3) the objects $a_!L_0$ then lie in $D^b_c(X, \mathcal{O}_E)$. Their reductions modulo π_E^n are bounded constructible complexes with coefficients in $\mathcal{O}_E/\pi_E^n$, which are pulled back from $X_{\acute{e}t}$ (Corollary 2.44*).

Proof of Lemma 3.25. In (1) and (2) we compare two coefficient sheaves related by a flat map that is an isomorphism modulo every π_E^n, so they have the same derived complete objects. For (3) we use the two characterizations of integral constructibility, while for (4) we combine the dévissage of Corollary 3.20 with the lattices of Lemma 3.24.

(1) and (2). The map from the constant sheaf $\mathcal{O}_E$ to $\mathcal{O}_{E,X}$ is flat and induces isomorphisms modulo every π_E^n [Stacks, 099L]. Hence by [Stacks, 099I] restriction of scalars is an equivalence on derived complete objects, which is (1). By [Stacks, 09C1], an object of $D_{\text{cons}}(X, \mathcal{O}_E)$ is a derived complete complex whose reduction modulo π_E is constructible of finite Tor dimension over $\mathbf{F}_E$. Since over the field $\mathbf{F}_E$ every bounded complex has finite Tor dimension, this description agrees with the description of $D^b_c(X, \mathcal{O}_E)$ in Proposition 3.11, and together with (1) this gives (2).

(3) By Proposition 3.11, an object of $D(X_{\text{pro\'et}}, \mathcal{O}_E)$ lies in $D^b_c(X, \mathcal{O}_E)$ if and only if it satisfies condition (2) there: it is derived complete with constructible reduction modulo π_E. Equivalently, it satisfies condition (3): there is a finite stratification such that on each stratum S, pro-étale locally, $K|_S \simeq \mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} P^\bullet$ for a bounded complex $P^\bullet$ of finite free $\mathcal{O}_E$-modules; here the implication from (2) to (3) is Proposition 3.10. For each of the three operations we therefore check whichever of these two conditions is convenient.

Inverse image. By Lemma 2.54*, f^* sends finite free $\mathcal{O}_{E,Y}$-modules to finite free $\mathcal{O}_{E,X}$-modules, so it preserves the local description of condition (3).

Extension by zero. The functor $a_! = i_* j_!$ preserves derived completeness. Indeed, $j_!$ for a quasi-compact open immersion j preserves limits by [Stacks, 09BP], and i_* is exact and preserves limits by [Stacks, 09AA, 09BM]. Moreover, both functors commute with reduction modulo π_E and preserve constructibility of the reductions, so $a_!$ preserves condition (2).

Tensor product. Let $K, M \in D^b_c(X, \mathcal{O}_E)$, and choose a stratification for K as above. Derived completeness and constructibility of the reduction modulo π_E are both preserved by cones, so $D^b_c(X, \mathcal{O}_E)$ is closed under cones. Moreover, the localization triangles along the strata exhibit K as a finite iterated extension of the objects $a_!a^*K$, for the strata $a : S \hookrightarrow X$, and the functor $- \otimes^{\mathbf{L}} M$ preserves these triangles. It therefore suffices to show that $(a_!a^*K) \otimes^{\mathbf{L}} M$ lies in $D^b_c(X, \mathcal{O}_E)$.

For this we use the projection formula for a locally closed immersion: for $L \in D(S_{\text{pro\'et}}, \mathcal{O}_E)$ and $M \in D(X_{\text{pro\'et}}, \mathcal{O}_E)$ the canonical map

$$(a_!L) \otimes^{\mathbf{L}} M \longrightarrow a_!(L \otimes^{\mathbf{L}} a^*M)$$

is an isomorphism. Indeed, write $a = i \circ j$ with j open and i closed, so that $a_! = i_* j_!$. On S the map restricts to the identity of $L \otimes^{\mathbf{L}} a^*M$, while on every complementary stratum both sides restrict to zero. Since restriction to the strata is exact and jointly conservative by [Stacks, 09AD], the map is an isomorphism.

Given the parts on inverse image and extension by zero, we are reduced to showing that $a^*K \otimes^{\mathbf{L}} a^*M$ lies in $D^b_c(S, \mathcal{O}_E)$. Tensoring with a^*K preserves derived completeness, because a^*K is pro-étale locally a bounded complex of finite free modules. Moreover, the reduction modulo π_E of the tensor product is the tensor product of the reductions, and this is constructible.

(4) For $\overline{\mathbf{Q}}_\ell$, we first descend to a finite E by Lemma 3.23. By Corollary 3.20, every object of $D^b_c(X, E)$ is obtained by finitely many cones and shifts from sheaves $a_! L$ with L lisse. We refine the stratum S by a finite stratification as in Lemma 3.24. This replaces $a_! L$ by finitely many extensions of such sheaves, and on each new stratum L contains a lattice L_0 with $L_0[1/\ell] = L$, so that $a_! L = (a_! L_0)[1/\ell]$. It remains to see that L_0 is integrally constructible. The reduction of L_0 modulo π_E^n is a locally constant sheaf of finitely presented $\mathcal{O}_E/\pi_E^n$-modules, so it comes from the étale site by [Stacks, 099Y]. Thus $L_0 \in D^b_c(S, \mathcal{O}_E)$, and $a_! L_0 \in D^b_c(X, \mathcal{O}_E)$ by (3). ∎

On $\mathbf{A}^1$, part (4) is visible in the running example. There the constant sheaf $E_{\mathbf{A}^1}$ is the cone of a map $i_* E[-1] \to j_! E_{\mathbf{G}_m}$, and up to shift both terms are of the form $(a_! L_0)[1/\ell]$ with L_0 the constant lattice $\mathcal{O}_E$ on the stratum.

3.10 Inverting ℓ

Passing from $\mathcal{O}_E$ to E means inverting ℓ, and on derived categories this is a Verdier localization.

Proposition 3.26 (Inverting ℓ on derived categories)**.** *Extension of scalars $E_X \otimes_{\mathcal{O}_{E,X}} - : D(X_{\text{proét}}, \mathcal{O}_E) \to D(X_{\text{proét}}, E)$ identifies $D(X_{\text{proét}}, E)$ with the Verdier quotient of $D(X_{\text{proét}}, \mathcal{O}_E)$ by the complexes K with $H^q(K)[1/\ell] = 0$ for all q.*

Proof. A functor with a fully faithful right adjoint is the Verdier localization at its kernel, so we show that restriction of scalars is fully faithful and compute the kernel.

Inverting ℓ is a filtered colimit along multiplication by ℓ, hence exact. Therefore $E_X = \mathcal{O}_{E,X}[1/\ell]$ is flat over $\mathcal{O}_{E,X}$, and $H^q(E_X \otimes K) = H^q(K)[1/\ell]$. Restriction of scalars is right adjoint to extension of scalars, and the counit is an isomorphism, because ℓ already acts invertibly on an E_X-module. Restriction is therefore fully faithful, and extension of scalars is the Verdier localization at its kernel. By the formula for $H^q(E_X \otimes K)$, this kernel consists of the complexes K with $H^q(K)[1/\ell] = 0$ for all q. ∎

The kernel contains more than the complexes killed by a power of ℓ, for example $\bigoplus_{n \geq 1} \mathcal{O}_{E,X}/\ell^n$. It does so even among constructible sheaves: on $\bigsqcup_{n \geq 1} \operatorname{Spec} k$, the sheaf equal to $\mathcal{O}_E/\ell^n$ on the n-th point is killed by inverting ℓ but by no power of ℓ. When the underlying space of X is Noetherian, however, this cannot happen in $D^b_c(X, \mathcal{O}_E)$, and there inverting ℓ amounts to tensoring the Hom groups with E.

Proposition 3.27 (Inverting ℓ)**.** *Assume that the underlying topological space of X is Noetherian (for instance, that X is of finite type over a field). A complex in $D^b_c(X, \mathcal{O}_E)$ that is killed by inverting ℓ is killed by a fixed power of ℓ, and extension of scalars induces an equivalence*

$$D^b_c(X, \mathcal{O}_E)[\ell^{-1}] \xrightarrow{\sim} D^b_c(X, E),$$

where the source category has the objects of $D^b_c(X, \mathcal{O}_E)$ and Hom groups tensored with E.

Proof. Both assertions are [BS15, Proposition 6.8.14(2)]. We prove the first, which is where the Noetherian hypothesis enters. Let $K \in D_c^b(X, \mathcal{O}_E)$. By Definition 3.1, there is a finite stratification such that the restriction of each of the finitely many cohomology sheaves of K to each stratum S is, on quasi-compact pro-étale charts U, of the form $\mathcal{O}_{E,U} \otimes_{\mathcal{O}_E} P$ with P finitely generated. Suppose that $K[1/\ell] = 0$. Extension of scalars to E_X is exact, so $E_U \otimes_{\mathcal{O}_E} P = 0$, and each such P is torsion. The underlying space of X is Noetherian, so each stratum is quasi-compact and finitely many charts suffice. With finitely many P, one power of ℓ kills the restrictions of all cohomology sheaves of K to all strata. By the localization sequences of Proposition 2.58*, a larger power kills every cohomology sheaf of K. Since K is bounded, a further power kills K itself, by induction along its truncation triangles.

Full faithfulness says that for $K, L \in D_c^b(X, \mathcal{O}_E)$ every morphism $E_X \otimes K \to E_X \otimes L$ is ℓ^{-c} times the image of a morphism $K \to L$ for some $c \geq 0$, and that a morphism $K \to L$ becomes zero after extension of scalars only if some power of ℓ kills it. For essential surjectivity, recall from Lemma lemma-integral-models that $D_c^b(X, E)$ is generated by objects coming from $D_c^b(X, \mathcal{O}_E)$. By the description of the Hom groups, a connecting morphism between them lifts to an integral one after multiplication by a power of ℓ. Hence finite cones of the generators also have integral models. ∎

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